# Air Quality Science: Lab

UTSC Event for Visiting Students

## Background

Monitoring and studying air quality involves measurements across a wide range of locations. There are measurement networks at every scale: global, national, regional and local. There are also satellites that acquire measurements at a variety of frequency and spatial sampling. In this lab, you will be using a relatively simple monitoring instrument to acquire measurements around University of Toronto's Scarborough (UTSC) campus.

The Temptop M2000+ sensor acquires measurements of:

- Particulate matter (PM) at 2.5 and 10 micrometers, and total particle count.
- $CO_2$
- HCHO (formaldehyde)
- Temperature and humidity

In an ideal situation, we would also like to acquire other measurements, such as ozone, $NO_2$, and CO. However, that would increase the size of the device, making it less mobile. It would also be much more expensive.

Background note: there are a variety of air quality measurements done in Toronto using other instruments – both at fixed locations and mobile. For example, a U of T group (led by Prof. Wunch) uses sensors attached to bikes. These are driven around the city by graduate and undergraduate students. This offers a balance of mobility, accurate measurements, and flexibility to reach many locations.

The device you will be provided was selected because PM2.5 is an important air quality quantity. It has a variety of sources, such as vehicle exhaust, industrial emissions, and other combustion (such as wildfires). It is typically combined with measurements of $NO_2$ and $O_3$ (ozone) to calculate an overall Air Quality Index. An AQI is calculated a variety of different ways around the world and using a variety of scales. The purpose is to summarize the exposure risk while outdoors and communicate this information to the public.

In addition, the device measures $CO_2$. There are two main reasons to monitor $CO_2$:

- $CO_2$ concentrations above ~1000 ppm can cause health impacts, e.g., headaches, decreased focus, and other symptoms.

- $CO_2$ is a well-mixed gas. As such, it is a proxy (tracer) for ventilation.

Finally, HCHO (formaldehyde) measurements can be made. HCHO is an air pollutant and respiratory irritant. It can be produced by the VOCs released by forests (e.g., isoprene). The sensor in the device is also cross-sensitive to other VOCs.

Lab Goals:

- Explore UTSC campus and acquire measurements at a variety of outdoor and indoor locations.
- Report back your observations to the class and discuss.

Specific Questions and Guidance

Before leaving, consider:

- Where do you expect to see high vs. low particulate matter?
- Where do you expect to see high vs. low $CO_2$?

During campus measurements:

- Where is the 'best' and where is the 'worst' air quality on campus? Start with your expectations (hypothesis) and compare to your measurements.
    - Note: I am not defining 'best' and 'worst': consider how you would define these terms and why.
- How does indoor $CO_2$ compare to outdoor $CO_2$? Are all indoor and outdoor areas the same? What accounts for the differences?
- Are there any patterns in PM2.5 levels? What factors seemed to influence your observations?
- How much did the values fluctuate while staying in one location of interest for a few minutes? What might explain this short-term variability?
- Decide in advance on one to three highlights you might want to discuss with the class once you return, e.g., an unexpected observation, a high-pollution (or low-pollution) location, or an indoor/outdoor difference.

After returning to lecture room and reflecting on your measurements:

- Did your observations about the 'best' and 'worst' locations for $CO_2$ and PM2.5 align with your expectations? Explain.
- Were there any surprising or contradictory results?
- If we examine the measurements of PM2.5 taken at the AA roof by a higher quality instrument, do you expect they will match your measurements?
    - What are possible reasons for discrepancies?
- If you were designing an air quality monitoring system for a city, how would you use these sensors, or sensors like it? How would you decide where to put them?